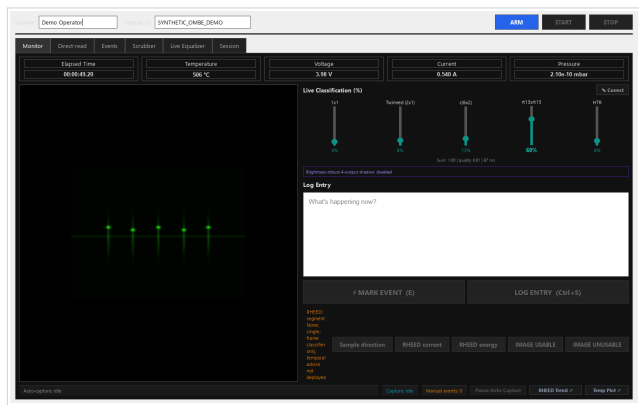
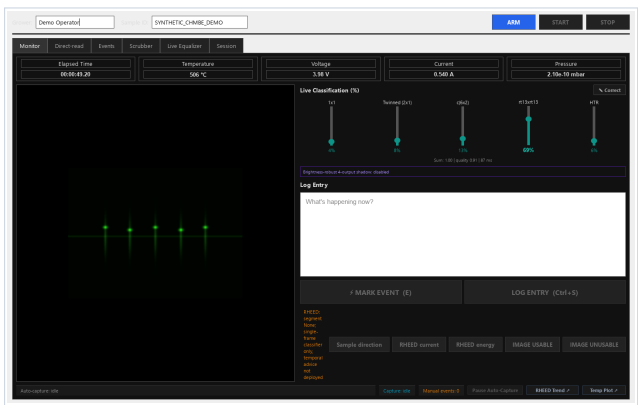


O-MBE + Ch-MBE GUI and RHEED Point-event Review

English Operator Manual



O-MBE Growth Monitor
Generated demo inputs; no hardware.



Ch-MBE Growth Monitor
Generated demo inputs; no hardware.

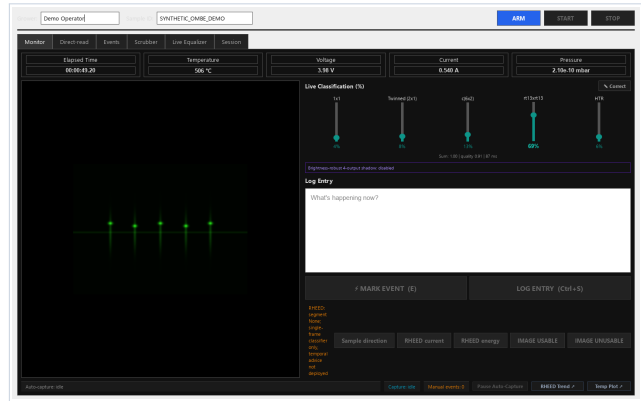
The post-processing labeler reads archived data only. It does not connect to or control cameras, pyrometers, power supplies, or any other instrument.

Quick entry	Double-click
O-MBE live GUI	Start O-MBE Growth Monitor.cmd
Ch-MBE live GUI	Start Ch-MBE Growth Monitor.cmd
RHEED offline labeler	Start RHEED Post-processing Labeler.cmd
Install or refresh desktop shortcuts	Install AI4MBE Desktop Shortcuts.cmd

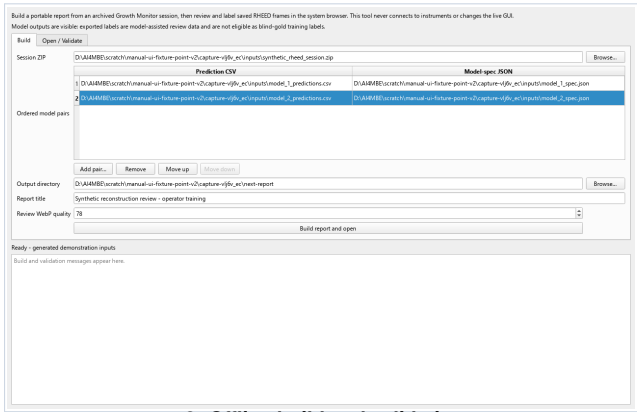
1 Understand the live and offline paths

In brief. The live GUI creates immutable session evidence. The offline labeler reviews manual, automatic-change, and posthoc point events. Equalizer remains separate from human interpretation and model output.

The live GUI performs acquisition and display. The offline labeler works only on completed archived sessions. Never interpret an offline result as a real-time control instruction.



1. Live acquisition
The chamber-selected Growth Monitor writes the archive.



2. Offline build and validation
The labeler combines the ZIP with predictions and model specifications.

The live GUI creates session records. The labeler reads those records later, produces a browser report, and exports validated JSON. It never sends commands back to the instruments.

What this manual covers

Chapter	Topic	Outcome
2	Chamber-specific live startup	Select the correct launcher and verify its exact title
3	Configuration modes and chamber defaults	Select the validated reader without cross-copying
4	Labeler desktop application	Select inputs, build, open, and validate
5-6	Inputs and safe report build	Produce a complete local report directory
7-8	Point events, Equalizer, and display controls	Complete traceable reviews on actual saved frames
9	Export, import, and validation	Produce canonical JSON and validate fail-closed
10-12	Safety, troubleshooting, and version checks	Handoff and reproduce the work safely

Human surface-reconstruction interpretation, Equalizer visual fit, image acquisition quality, model output, and FeSe film quality are different concepts. Image acquisition quality means only whether a captured frame can be analyzed; it never means poor surface or film quality. The current classifier concerns the bare STO surface before growth.

Operating principle

- The selected O-MBE or Ch-MBE GUI reads live instrument sources and writes chamber-identified session records under the operator’s control.
- Post-processing combines an archived session with one or more already-generated prediction tables.
- Reviews are stored as frame-bound point events with immutable source evidence and append-only revisions.
- Equalizer is an auxiliary four-basis visual fit; it never fills or overwrites a human reconstruction field.
- Model outputs remain visible, so these annotations are assisted review, not blind-gold labels.

2 Install shortcuts and start the correct chamber GUI

In brief. Choose exactly one chamber launcher, verify its exact window title, and check live status before ARM. The launchers isolate chamber identity, but they do not authorize operation or setpoint changes.

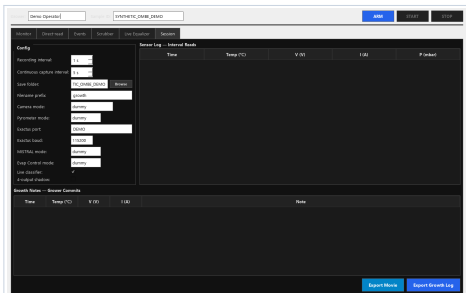
Shared first setup or after moving the checkout

1. From the GUI repository root, double-click `Install AI4MBE Desktop Shortcuts.cmd`.
2. Confirm that the desktop contains **O-MBE Growth Monitor**, **Ch-MBE Growth Monitor**, and **RHEED Post-processing Labeler**.
3. If Windows displays an unexpected security warning, stop and ask the maintainer to verify the file source, branch, and commit. Do not bypass it.

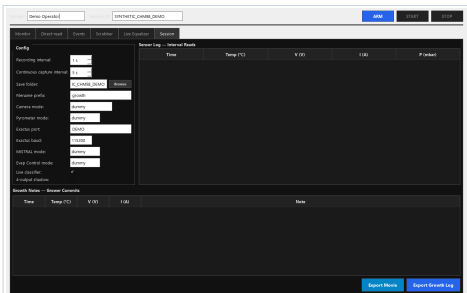
Select exactly one chamber branch

Physical chamber	Required launcher	Exact window title
O-MBE	Start O-MBE Growth Monitor.cmd	Oxide MBE Growth Monitor
Ch-MBE	Start Ch-MBE Growth Monitor.cmd	Chalcogenide MBE Growth Monitor

A different title or a launcher that does not match the physical chamber is a stop condition. Close normally and restart with the correct launcher; changing a GUI field is not a substitute.



O-MBE Session tab. Generated demo inputs; not operating defaults.



Ch-MBE Session tab. Generated demo inputs; not operating defaults.

Shared ARM and exit boundary

1. Confirm the title-selected chamber, advancing RHEED sequence, valid temperature state, and intended log directory. Check the four reader modes against the matching startup-default record in Chapter 3; verify all remaining operating settings against the applicable SOP.
2. ARM or start only when the authorized operator and applicable SOP permit it. Double-click startup authorizes no instrument operation or setpoint change.
3. Close normally and wait for the window to disappear. Do not force-kill Python while logs are being written.

Each launcher forces and preflights its chamber profile and owns a chamber-specific single-instance mutex. The selected Python must import `vmcpy`, `pyads`, and `serial`; O-MBE must also import `pymodbus`. A missing production dependency blocks startup instead of opening a GUI whose default readers cannot work. Logs below `%LOCALAPPDATA%\AI4MBE\LauncherLogs` record Python, repository, branch, commit, chamber, required modules, and dependency result. Windows Graphics Capture is an optional diagnostic mode and is checked only when selected. Do not create global environment variables ad hoc.

3 Understand Configuration modes and chamber defaults

In brief. Both chambers start with vimba, modbus, ads, and elog: direct reads for RHEED, temperature, MISTRAL, and EvapControl. dummy modes generate test data and must not be mistaken for instruments.

The Configuration panel chooses one reader for each data source. A mode selects *how the GUI reads*; it does not change an instrument setpoint. The four selectors are independent, and their selected values are frozen when the session is armed.

Camera mode - RHEED image source

Option	Meaning
dummy	Fixed generated/demo STO 1x1 image; no camera connection.
dummy_c6x2	Fixed c(6x2) demonstration image; no camera connection.
dummy_tw	Fixed Twinned (2x1) demonstration image; no camera connection.
dummy_rt13_tilted	Rotated RT13 demonstration image for alignment testing; no camera connection.
screengrab	Windows Graphics Capture of a detached kSA <i>Live Video</i> window. It captures the window even when another window covers it, but it is still display pixels rather than raw camera data.
screengrab_mss	Legacy monitor-pixel capture. Window position, overlap, DPI, and foreground order can contaminate the frame; diagnostic fallback only.
vimba	Direct AVT camera acquisition through the Vimba SDK. This is the default live RHEED source. kSA camera ownership/access mode must permit the connection.

Pyrometer mode - substrate temperature source

Option	Meaning
dummy	Generated temperature values for offline GUI checks; not an instrument reading.
exactus	Direct serial read using the Exactus protocol and the displayed COM/ baud fields.
modbus	Direct read from the pyrometer using the chamber profile's COM, baud, RTS, device ID, and Modbus backend. This is the default.
screengrab	Reads the existing TemperaSure application through Windows UI automation. It depends on the correct window being open and updating.

MISTRAL mode - power and cell data source

Option	Meaning
dummy	Generated voltage/current values; no PLC or MISTRAL connection.
screengrab	Captures the MistralGui window and OCRs setpoint and actual voltage/current. The window must be visible and the calibrated layout must match.
jsonrpc	Experimental HTTP direct connection. Its read-method map is not configured, so it can connect while returning no V/I values; do not use for production logging.
ads	Read-only direct Beckhoff TwinCAT ADS access using the chamber-specific endpoint, ports, schema, and cell count. This is the default.

EvapControl mode - pressure and evaporation data source

Option	Meaning
dummy	Generated pressure values; no EvapControl connection.
elog	Directly reads EvapControl's current .elo binary log. It needs no OCR or visible window and can expose pressure, substrate, cell, and plasma fields present in the validated schema.
screengrab	Captures the Evaporation Control window and OCRs chamber pressure. The window must be visible; fields not shown in that crop remain unavailable.

vimba, modbus, ads, and elog avoid screen OCR, but each worker still receives data independently. A sensor-log row is a latest-value software snapshot, not proof that exposure, temperature, PLC, and EvapControl were physically sampled at one instant.

O-MBE GUI startup defaults

Selector	Startup selection and source
Camera	vimba - direct AVT camera read
Pyrometer	modbus - COM4, 115200 baud, device 1, RTS off, pymodbus backend
MISTRAL	ads - chamber-specific read-only ADS profile (6 cells)
EvapControl	eelog - direct current .eelog read

Ch-MBE GUI startup defaults

Selector	Startup selection and source
Camera	vimba - direct AVT camera read
Pyrometer	modbus - COM3, 115200 baud, device 1, RTS off, raw_serial backend
MISTRAL	ads - chamber-specific read-only ADS profile (7 cells)
EvapControl	eelog - direct read from the Ch-MBE .eelog log directory

These are software startup selections, not permission to ARM. ROI/crop, classifier package and enable state, save-root approval and naming, sampling intervals, and any instrument setpoint remain governed by the matching chamber SOP and owner approval. The initial Windows save field uses separate OMBE and ChMBE folders and remains editable before ARM. Each production profile passes its own explicit directory to ElogReader; the generic AIQM_EVAP_LOG_DIR remains a standalone-reader fallback and cannot override the GUI profile. Session metadata records the effective pyrometer serial fields, configured Elog directory, and exact .eelog source once connected. Variables absent from that chamber's schema remain blank rather than being invented.

4 Open the RHEED post-processing labeler

In brief. Use the desktop labeler when Equalizer or Complete is required. It remains offline from instruments and exposes only a random-token service on 127.0.0.1 for the selected local report.

Step 1. Double-click Start RHEED Post-processing Labeler.cmd or its desktop shortcut.

Step 2. Confirm that the application has the **Build** and **Open / Validate** tabs, plus **Build report and open**, **Open report**, **Validate annotations**, and **Open English PDF manual**. It must not start Growth Monitor or contact instruments.

Build a portable report from an archived Growth Monitor session, then review and label saved RHEED frames in the system browser. This tool never connects to instruments or changes the live GUI. Model outputs are visible: exported labels are model-assisted review data and are not eligible as blind-gold training labels.

Build Open / Validate

Session ZIP D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\inputs\synthetic_rheed_session.zip Browse...

	Prediction CSV	Model-spec JSON
1	D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\inputs\model_1_predictions.csv	D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\inputs\model_1_spec.json
2	D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\inputs\model_2_predictions.csv	D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\inputs\model_2_spec.json

Ordered model pairs

Add pair... Remove Move up Move down

Output directory D:\AI4MBE\scratch\manual-ui-fixture-point-v2\capture-vj6v_ec\next-report Browse...

Report title Synthetic reconstruction review - operator training

Review WebP quality 78

Build report and open

Ready - generated demonstration inputs

Build and validation messages appear here.

Actual application screenshot. The current desktop labeler Build tab populated with a generated local fixture. The application is offline and no instrument process is running.

Area	Purpose
Session ZIP	Select one archived Growth Monitor session
Model inputs	Pair one prediction CSV with its model-spec JSON on each row
Output directory	Select a new or empty directory, preferably outside Git
Message log	Preserve complete build and validation messages
Open report button	Open a previously generated interactive_report.html
Validate annotations	Check exported JSON against the exact report provenance

This application can run on a computer without instrument software when the archived inputs are complete and the Python environment is valid. Report construction and validation run in an isolated child process so the desktop application remains responsive.

The desktop application serves the selected report only on the Windows loopback interface and gives it a random session token. The browser can then request the existing retrospective PyQt Equalizer and audited revisions. A report opened directly with file:/// can navigate and edit Draft fields, but **Run Equalizer** and **Complete** remain unavailable and explain that the desktop labeler is required.

5 Prepare inputs and verify provenance

In brief. Preserve the ZIP and exact saved frames. Manual marks and automatic image-change points become editable Drafts; adjustment, image-unusable, and instrument records remain read-only references.

Session archive

The ZIP archive must contain exactly one member ending in `session_metadata.json`, exactly one member ending in `heartbeat_log.csv`, and every sibling frames/ image referenced by the heartbeat rows. Saved-frame elapsed time, heartbeat index, capture sequence, and timezone-aware capture UTC must each increase strictly; gaps are allowed. Do not extract and rename frames manually.

Read `chamber_id` from the session metadata and confirm that it names the intended O-MBE or Ch-MBE source. For Modbus, preserve COM, baud, RTS, device ID, and backend. For ADS, preserve endpoint, ports, and cell count. For Elog, preserve the configured directory and resolved source file. Compare provenance only with the corresponding startup-default record in Chapter 3; never edit an archive to make it resemble the other chamber.

```
growth_session.zip
  session_metadata.json
  heartbeat_log.csv
  frames/
    heartbeat_000001_....bmp
    heartbeat_000002_....bmp
```

Prediction tables and model specifications are positional pairs

File	Required content	Check
Prediction CSV	frame index, heartbeat index, elapsed time, capture UTC, capture sequence, frame name, frame SHA-256, and score columns	Rows match the saved-frame order exactly
Model-spec JSON	schema version 1, non-empty key and title, at least two unique classes, and equally sized unique probability columns	Class order matches the score columns; status and provenance are optional
Session ZIP	metadata, heartbeat log, and frames	Every referenced frame exists in the archive

Prediction `frame_index` may begin at 0 or 1, but it must then remain contiguous. Heartbeat indices, capture sequences, and elapsed times may contain gaps. The tool follows actual saved frames and recorded times; it never assumes exact 1 Hz sampling.

Any mismatch in row count, order, time, sequence, filename, or SHA-256 stops the build. Do not edit the inputs to bypass an error.

Event and context sources

- `manual`: one-click **MARK EVENT** evidence created during acquisition; an empty comment remains Unfinished.
- `auto_capture`: a point produced by the RHEED image-change detector. It marks a visual change, not its physical cause.
- `posthoc`: a point added later at an actual saved frame.
- RHEED direction, beam-current, and beam-energy adjustments are read-only reference points.
- An image-unusable record means only that acquisition or capture made the image unsuitable for analysis. It is not a surface-quality or film-quality judgment.
- Temperature, voltage, current, pressure, and data age come from original logs and remain read-only context, not editable label fields.

6 Build and open a report safely

In brief. Build only into a new or empty directory outside Git. Preserve the complete report directory and the original prediction/model-spec inputs separately.

Older events_labels.csv rows may seed automatic-event review. A legacy live_labels.csv row links only when capture sequence and frame SHA-256 identify exactly one event. Legacy temporal segments remain readable but are not converted automatically because a segment has no unique scientific point.

Recommended local layout

```
<local-data-root>\2026-08-11_anneal\
  source\growth_session.zip
  predictions\model_A.csv
  specs\model_A.json
  output\
  exports\
```

Step 1. Choose the preserved archive copy in Session ZIP.

Step 2. Add one Prediction CSV and Model-spec JSON row for every model to display. Pairing is positional, so preserve row order.

Step 3. Choose a new or empty directory outside the repository. The desktop labeler will not overwrite a non-empty directory.

Step 4. Enter a report title. Review quality changes only the lossy review images; it never changes archived pixels or model predictions. The default value of 78 is suitable for routine use.

Step 5. Click **Build report and open**. Wait for success and for interactive_report.html to open automatically.

PowerShell fallback

```
Set-Location '<GUI repository>'
conda activate ai4mbe-gui
python -m tools.rheed_postprocessing_labeling build `
  --session '<local-data-root>\growth_session.zip' `
  --predictions '<local-data-root>\predictions.csv' `
  --model-spec '<local-data-root>\model_spec.json' `
  --output-dir '<local-data-root>\point-event-review'
```

Copy the entire report directory

The report consists of interactive_report.html, images/, vendor/, run_manifest.json, and its adjacent annotations/ sidebar. Copy the entire directory during handoff. Preserve the original prediction CSV and model-spec JSON separately outside the report directory. Review WebP files are display copies only; Equalizer reads the original BMP or PNG bytes from the read-only ZIP.

The desktop application always refuses a non-empty output directory. CLI --overwrite may replace only a recognized, unmodified tool output. It validates the new inputs and stages a complete replacement first. If the old report changes during the build, replacement stops and the original content is restored. Filesystem roots, repository paths, the current directory, and directories containing inputs are always rejected.

7 Review and complete point events

In brief. Select an Unfinished point, snap its review anchor to the best saved frame, add comment and reviewer, run Equalizer, and explicitly click Complete. There are no Mark In or Mark Out interval controls.

The rheed-point-events-v1 schema has three editable sources: manual, auto_capture, and posthoc. Multiple events may share one timestamp or frame while keeping different IDs. Current events use UUIDs; legacy IDs deterministically include session, source file, source index, time, and capture sequence.

Immutable original point and editable review point

The original point retains event ID, source, time, source frame when available, capture sequence, image SHA-256, original note, and source-row SHA-256. It never moves. The review point can move only by snapping to an actual saved frame. Moving it clears the old Equalizer measurement because that result belonged to a different image.

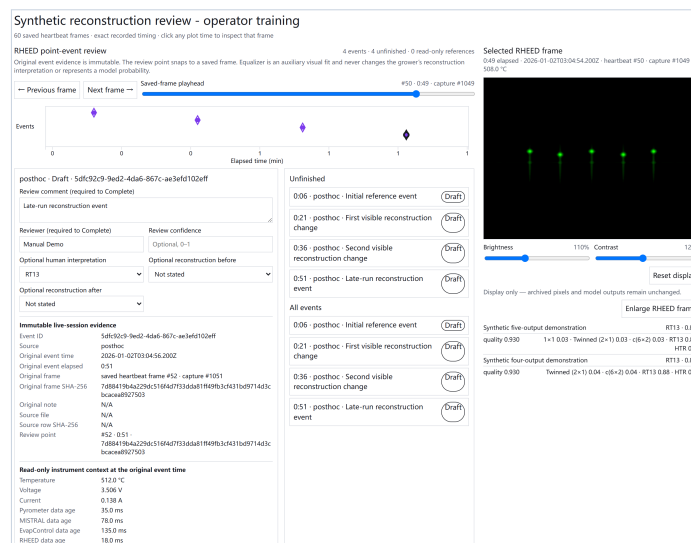
Step 1. Select an event in **Unfinished**, or choose **Add posthoc event at playhead** on an actual saved frame.

Step 2. Compare the immutable original marker with the movable review marker. Move only when another saved frame is a better review anchor.

Step 3. Enter a meaningful comment and reviewer. Confidence and human reconstruction before/after fields are optional.

Step 4. Click **Run Equalizer**; align and confirm handedness when no compatible accepted calibration exists.

Step 5. Adjust and save Equalizer, save the Draft, and click **Complete** explicitly.



Actual application screenshot. The actual point-event editor using generated demo inputs. Point markers and the Unfinished queue replace the former interval-label controls.

Complete requires a nonempty comment, reviewer, saved review frame, and valid Equalizer result for that exact frame. Editing a completed event's comment, review point, or Equalizer returns it to Draft. Use **Reopen** for intentional further review.

A source event from manual or auto_capture cannot be deleted. It may be dismissed only with a reason, preserving the evidence. A posthoc event can be deleted and restored through audited revisions.

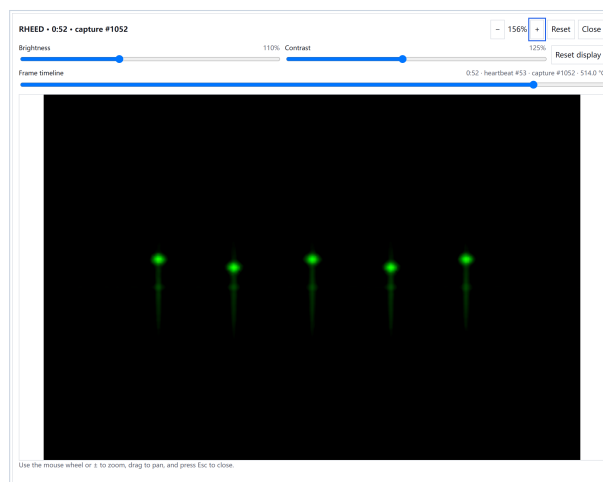
8 Interpret Equalizer, logs, and display controls

In brief. Equalizer is a separate visual fit, sensor values are read-only context, and brightness, contrast, zoom, and enlargement never modify archived pixels or model outputs.

Equalizer runs only for the selected event. It records calibration and basis-bundle IDs, raw/final/normalized weights, fit residual, valid coverage, and the exact frame hash. The active bases are 1x1, Tw, c6x2, and RT13. HTR has no canonical basis and must remain null.

Equalizer weights are a visual linear fit. They are not model probabilities, area fractions, or a human reconstruction judgment. Equalizer never fills or overwrites a human reconstruction field, and model argmax never overwrites Equalizer or human review.

When no compatible calibration exists for the selected frame, view segment, geometry, and basis hash, complete the existing three-point alignment and handedness confirmation. Equalizer reads the original BMP or PNG from the ZIP, not a lossy WebP preview.



Actual application screenshot. The actual enlarged-frame dialog. Its timeline changes the selected saved frame while preserving zoom, pan, brightness, and contrast.

The image, playhead, model guides, point markers, and enlarged timeline remain synchronized. Temperature, voltage, current, pressure, and data age are looked up from the original logs at event time and remain read-only. A common displayed time does not prove physically simultaneous sampling.

9 Preserve revisions, drafts, and exports

In brief. Every edit is append-only. The source ZIP and acquisition CSVs remain unchanged, offline changes live in annotations/, and canonical JSON retains more provenance than CSV.

Every revision records actor, UTC, action, complete before/after state, and base revision ID. Live sessions write `rheed_event_revisions.jsonl` plus a rebuildable current-state summary. Offline review writes an adjacent sidecar; it never changes the ZIP. A transaction marker permits exactly-once recovery after interruption.

Browser local storage is only a convenience Draft. Export JSON frequently. CSV is useful for tables and collaboration but does not replace nested event, Equalizer, and revision provenance in JSON.

```
python -m tools.rheed_postprocessing_labeling validate `
--report '<local-data-root>\point-event-review\interactive_report.html' `
--annotations '<local-data-root>\point-event-review\annotations\point_events.json'
```

Validation binds the archive, ordered frame hashes, model context, event IDs, immutable source evidence, review anchors, revision chain, Complete gate, and Equalizer frame hash. Never bypass a mismatch. Old `rheed-temporal-segments-v1` files remain available to their original validator as read-only compatibility data; do not convert segment start, middle, or end into a point automatically.

10 Data safety and scientific boundaries

In brief. Keep real laboratory data and outputs out of Git. This workflow is model-assisted review of bare STO reconstruction, not blind-gold truth, acquisition-quality labeling, or FeSe film-quality evaluation.

May be tracked	Keep out by default
Tool code, templates, tests, and documentation	Real ZIP files, raw RHEED images, and sensor logs
Example model specifications	Predictions, generated reports, annotations, and check-points
Sanitized screenshots made with generated inputs	Credentials, unpublished results, and experiment screen-shots

Model plots and Equalizer are visible during this workflow, so its reviews are model-assisted and not blind gold. Image acquisition quality means whether a captured frame can be analyzed. It is never evidence of poor surface or film quality. The current reconstruction target is bare STO before growth, not FeSe film quality.

Preserve the read-only source ZIP and SHA-256, report manifest, revision journal, canonical JSON, model specifications, and complete report directory during handoff.

11 Troubleshoot without destroying evidence

In brief. Record the exact error, time, chamber, launcher, commit, environment, paths, event ID, and validity state before changing anything. Never copy settings between chambers or hand-edit evidence.

Symptom	Safe action
Wrong chamber title	Close normally and use the matching chamber launcher
Instrument value invalid	Record mode, connected/valid/error state, sequence, and age; do not guess COM or set-point values
Build rejects inputs	Check ZIP, prediction/spec pairs, hashes, and an empty output directory
HTML has no images	Restore the complete report directory
Unfinished event missing	Verify source CSV row and row hash; preserve revision replay errors
Run Equalizer unavailable	Reopen through the desktop labeler, not <code>file:///</code>
Equalizer rejected	Verify raw frame, accepted calibration, handedness, geometry, view segment, and basis hash
Complete disabled	Supply comment, reviewer, saved review frame, and valid exact-frame Equalizer
Revision recovery fails	Preserve JSONL and the pending marker; do not edit either file

For O-MBE, confirm **Oxide MBE Growth Monitor** and use only the O-MBE startup record. For Ch-MBE, confirm **Chalcogenide MBE Growth Monitor** and use only the Ch-MBE startup record. Preserve `git status --short --branch`, `git rev-parse HEAD`, launcher logs, and full validation messages.

12 Verify versions and finish the checklist

In brief. Record versions and hashes, complete only the selected chamber checklist, resolve or dismiss every Unfinished event, and validate the canonical point-event export.

```
Set-Location '<GUI repository>'
git status --short --branch
git rev-parse HEAD
conda activate ai4mbe-gui
python --version
python -m tools.rheed_postprocessing_labeling --help
Get-FileHash '<local-data-root>\growth_session.zip' -Algorithm SHA256
```

O-MBE startup checklist

- ☐ Use Start O-MBE Growth Monitor.cmd and confirm **Oxide MBE Growth Monitor**.
- ☐ Confirm launcher repository, branch, commit, Python, and required production-driver probe.
- ☐ Confirm vimba / modbus / ads / elog against the O-MBE record.
- ☐ Confirm advancing RHEED, valid instrument states, data age, and log directory.
- ☐ ARM only under the O-MBE SOP and authorized operator.

Ch-MBE startup checklist

- ☐ Use Start Ch-MBE Growth Monitor.cmd and confirm **Chalcogenide MBE Growth Monitor**.
- ☐ Confirm launcher repository, branch, commit, Python, and required production-driver probe.
- ☐ Confirm vimba / modbus / ads / elog against the Ch-MBE record.
- ☐ Confirm advancing RHEED, valid instrument states, data age, and log directory.
- ☐ ARM only under the Ch-MBE SOP and authorized operator.

Shared offline checklist

- ☐ Confirm session metadata names the intended chamber and preserve the ZIP hash.
- ☐ Open through the desktop labeler when Equalizer or Complete is required.
- ☐ Resolve every Unfinished item or dismiss it with a documented reason.
- ☐ Confirm each Complete event has comment, reviewer, exact-frame Equalizer, and revision history.
- ☐ Export JSON, validate it, and hand off the complete report directory.

Searchable manual: docs/RHEED_GUI_Postprocessing_Labeling_User_Manual_EN.md

AI prompt pack: docs/RHEED_GUI_Postprocessing_Labeling_AI_Prompt_Pack_EN.md

Stop and contact the maintainer when instrument state, stale data, version identity, provenance, event recovery, or validation cannot be explained.